

Fatigue analysis of segmental precast post-tensioned concrete towers under operational wind turbine loads

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ARTICLE INFO

Keywords:

Fatigue calculation
Wind turbine
Concrete
Precast
Post-tensioned

ABSTRACT

Recently, segmental precast post-tensioned concrete towers have emerged as a solution for achieving higher hub heights in wind turbines. They are commonly employed as the lower portion of a hybrid tower, with a tubular steel tower situated on top. In this paper, fatigue analyses and evaluations of this tower using a multi-body dynamic model are presented. Initially, the key dimensions of a 160-m hybrid tower were determined based on operational limit state loads. The fatigue life calculations of the segmental precast post-tensioned concrete tower were analysed using this computational model. The influences of the mean stress correction, initial prestress, azimuth angle, and sample size were investigated. The results showed that fatigue accumulation in the horizontal joints of the segmental precast post-tensioned concrete tower was notably influenced by fatigue cycles with larger mean and range values. Furthermore, structural uncertainty analyses revealed that the elastic modulus of the concrete, tower foundation stiffness, and turbine control strategy significantly affected the fatigue damage of the segmental precast post-tensioned concrete tower.

1. Introduction

Wind turbine towers are subjected to alternating stress cycles owing to the variability in the turbulent wind input and turbine control system [1,2]. The accumulated fatigue damage over the lifetime of a tower often governs its design criteria. Fatigue stress calculation involves various fully coupled dynamic simulations under normal turbine operational conditions [3]. This task can be highly time-consuming for the fatigue design of a tower [4,5], particularly considering that a wind turbine is expected to operate for 20–50 years. The total number of fatigue stress cycles can reach approximately 10^9 [6]. To streamline computational efforts in practice, tower lifetime damage is typically estimated using a collection of stress time histories covering a shorter time range than the intended lifespan [7–10]. The duration of the stress time histories must be sufficiently long to ensure the statistical reliability of the fatigue stresses.

The fatigue life estimation accuracy is often evaluated using different year-long databases of dynamic responses [11–13]. Tubular steel towers are the primary choice for supporting wind turbines at a height of 80 m above ground over the past few decades [14–16]. This has been the focus of the previous studies on the fatigue analyses of wind turbine towers.

Moriarty [13] simulated five years of wind turbine loads during operation to facilitate load extrapolation analyses of tubular steel towers. Recently, steel-concrete hybrid towers have been proposed for onshore wind turbines to capture wind resources at greater heights [17–19]. Hybrid towers utilise segmental precast post-tensioned (SPPT) concrete towers as partial replacements for tubular steel towers, addressing cost barriers associated with large steel sections, as shown in Fig. 1. It is estimated that more than 5000 wind turbines in China are currently supported by hybrid towers. Although various concrete tower products have been introduced in the wind-energy industry, there are currently no established regulations for their fatigue design.

Previous research [20–24] has highlighted significant differences in the fatigue behavior of steel and concrete materials, indicating that the fatigue analysis of SPPT towers requires further investigation. Therefore, this study aims to assess whether additional factors should be incorporated into the fatigue calculations of an SPPT tower under operational wind turbine loads. The work presented in this paper is mainly divided into three parts. First, the multi-body dynamic model developed to evaluate the fatigue damage of an SPPT tower is described. Second, the key design parameters were studied to reveal their effects on the SPPT tower fatigue analyses. Finally, uncertainty analyses were

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<https://doi.org/10.1016/j.engstruct.2025.120295>

Received 8 August 2024; Received in revised form 18 December 2024; Accepted 9 April 2025

Available online 14 April 2025

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Fig. 1. On-site photos of the hybrid tower in China.

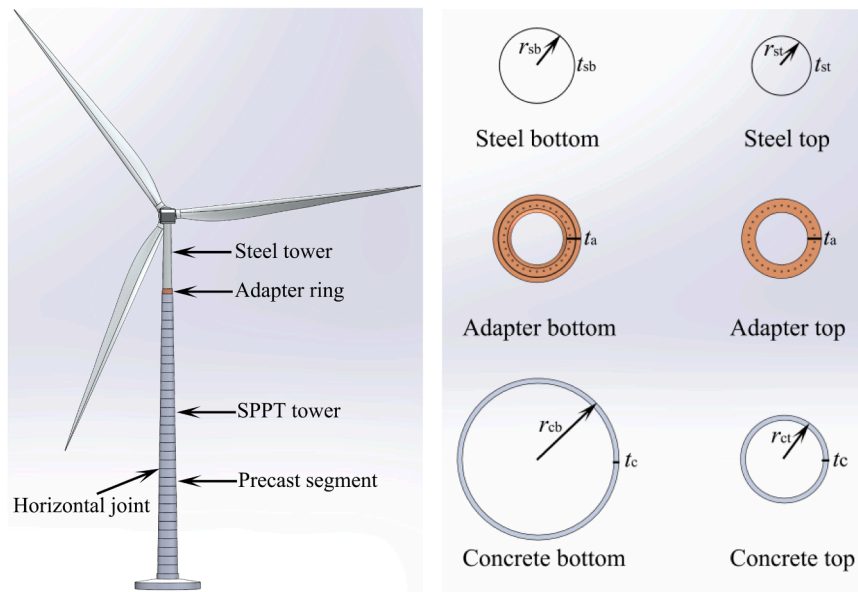


Fig. 2. Primary components of the steel-concrete hybrid tower.

performed to quantify the correlation between project-specific factors and the SPPT tower fatigue damage, including the elastic modulus of concrete, tower foundation stiffness, and turbine control strategy.

2. Description of the SPPT tower

Fig. 2 shows the layout of the primary components of the land-based SHT. The tubular steel tower is erected atop the SPPT tower with an interposed adapter ring to facilitate force transfer between the two substructures. Both tower portions have circular hollow sections with diameters that decrease linearly with height, although the inclination angles of two portions may vary slightly. The tapered geometry reflects the current trend in the wind energy industry. In Fig. 2, r_{cb} and r_{ct} represent the radii of the bottom and top sections of the concrete portion, respectively; t_c and t_a denote the thicknesses of the concrete portion and the adapter ring; r_{sb} and t_{sb} denote the radius and thickness of the bottom steel section; r_{st} and t_{st} are the radius and thickness of the top steel section.

The SPPT tower is constructed by stacking discrete segments on top

of one another. To accelerate the tower construction on site, these segments are typically designed without reinforcement across the joints and are clamped together using post-tensioned (PT) unbonded strands, as shown in Fig. 3. The PT strands are anchored at the cast-in-place foundations and the apex of the adapter ring, resulting in extremely long unbonded strands along the tower height; for instance, it can typically exceed 100 m in a 160-m SHT. Such a length prevents the strand from producing significant force variations owing to lateral deformations, which are often observed in precast bridge piers or shear walls with much smaller heights (e.g. [25,26]). Furthermore, SPPT towers typically undergo small lateral deformations under operational wind turbine loads [27], thereby causing fatigue stresses in strands to fluctuate within a very narrow range.

In each precast segment, longitudinal reinforcement terminates before reaching its top and bottom interfaces, leaving the PT strands as the main connectors across the contact interfaces between adjacent segments, which are typically termed horizontal joints. The lateral resistance of these joints is dependent on the pre-stressing force and the roughness of the interfaces. These locations typically experience greater

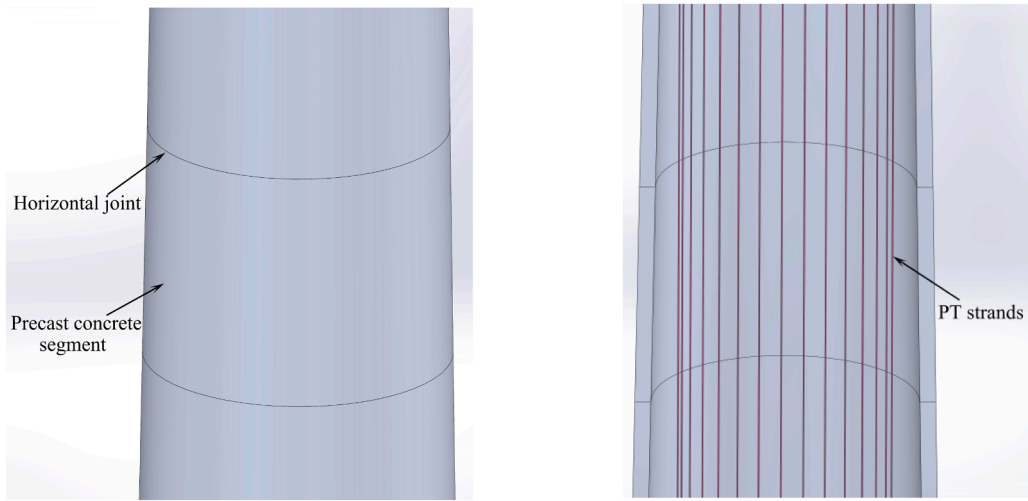


Fig. 3. External and internal views of the SPPT tower.

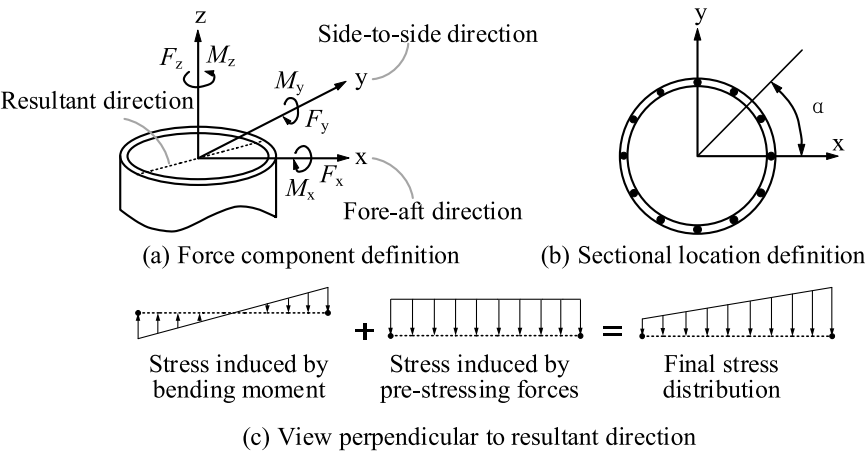


Fig. 4. Conceptual drawing showing the sectional stress used in fatigue damage evaluation.

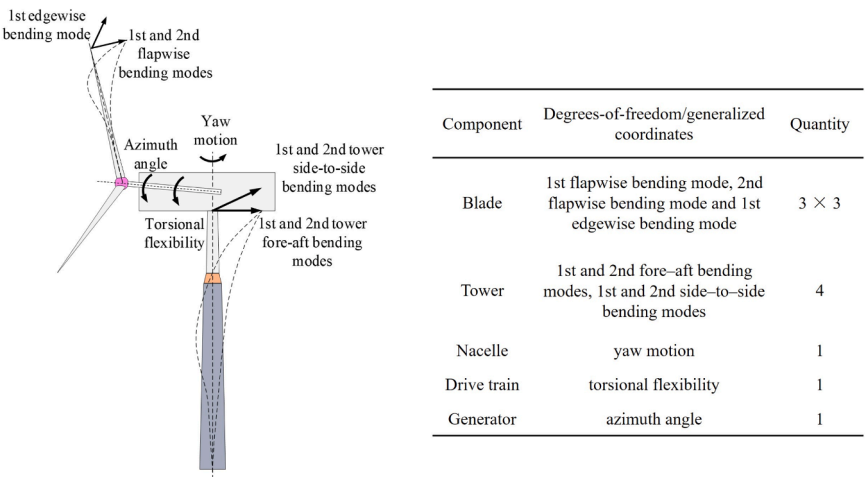


Fig. 5. Degrees-of-freedom of different components in the model.

stress variations compared to the segment bodies with reinforcements, making them critical for fatigue design in SPPT tower.

Under operational wind turbine loads, the horizontal joints experience different force components in both the fore-aft and side-to-side

directions, as illustrated in Figs. 4a and 4b. For tall and slender structures primarily subjected to top loadings, flexural responses are more critical than shear demands. Therefore, this study focuses on the fatigue damage induced by normal stresses, with the assumption that friction is

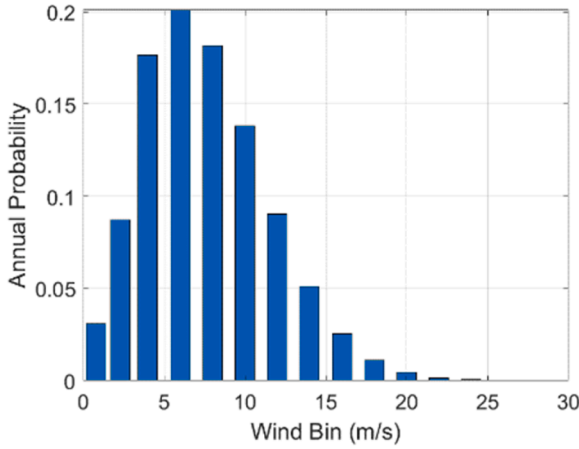


Fig. 6. Annual probability of wind speeds.

sufficient to prevent sliding throughout the planned lifetime. In Fig. 4c, the stress distribution resulting from the combination of two orthogonal bending moments M_x and M_y is presented. With the addition of pre-stressing forces, the horizontal joints are capable of preventing the development of tensile stresses during normal turbine operation.

3. Multi-body dynamic model

To simulate the operational turbine responses, the wind turbine generator system—comprising the blade, tower, nacelle, drivetrain, and generator—was idealised as a multi-body dynamic model using Kane's method in OpenFast software [28]. A standard ROSCO controller [29] was used for the turbine control system, which generated the primary source of nonlinearity in the model. The complete kinematics of the model was described by 16 degrees of freedom (DOFs), as shown in Fig. 5. Dynamic analyses were conducted with a time step of 0.005 s. Estimating the fatigue life of the wind turbine generator system requires numerous simulations following design load case (DLC) 1.2 in IEC 61400-1 [3].

The SCHT was designed to support an NREL 5-MW wind turbine with a hub height of 160 m at an IEC Class-III wind site [3]. The tower was assumed to be rigidly fixed to the ground and designed to have a minimum lifespan of 50 years. The annual mean wind speed at the hub

height V_{ave} was 7.5 m/s, and the reference turbulence intensity (I_{ref}) was 0.16. The hub-height mean wind speed V_{ave} follows a Rayleigh probability distribution, as detailed in the Appendix. The wind speeds were binned according to the V_{ave} distribution, each with an interval of 1 m/s, as shown in Fig. 6. The majority of the wind bins were clustered around the annual mean wind speed, with higher and lower wind intensities showing markedly lower occurrence probabilities. For the NREL 5-MW wind turbine, the cut-in and -out wind speeds were 3 and 25 m/s, respectively, corresponding to 11 wind bins.

Wind-field simulations are crucial for the fatigue analysis of wind turbine systems [30]. In this study, the wind field acting on the wind turbine was represented by a 19×19 grid covering an area of $170 \times 170 \text{ m}^2$. The mean wind speed profile was determined using a power law with an exponent of 0.2. The wind direction was assumed to be constant. The 3-D wind speed field (i.e., the x-, y-, and z-components along the x-, y-, and z-directions, as shown in Fig. 7) was generated using spectral representation methods (SRMs). An example of the x-, y-, and z-component generation at the hub height is shown in Fig. 7. The wind loading duration was extended by an additional 50 s to eliminate the effect of the initial conditions on the load statistics. The Kaimal spectrum [3] used to define the single-point power spectral densities (PSDs) was given in the Appendix. The coherence function was modelled using the exponential coherence model. The longitudinal wind speed (i.e., the u-component) comprises both the mean and fluctuating wind components. The non-Gaussian characteristics of the fluctuating wind components were not simulated.

The geometric parameters of the SCHT were initially determined using a genetic algorithm. This procedure was described by the authors of this paper in previous study [18]. The SPPT and steel towers were made of C70 concrete and S355 steel, respectively, in accordance with Eurocode [31,32]. The former has an elastic modulus of 41,000 MPa and density of 2500 kg/m^3 . Correspondingly, those of the latter were 210,000 MPa and 7850 kg/m^3 , respectively. Both materials were modelled as elastic uniaxial materials in OpenFast software. The ultimate limit state (ULS) and fatigue limit state (FLS) of the trial design were then rechecked using a group of 10-min dynamic load analyses. Extreme load extrapolation in the ULS was performed using a Weibull three-parameter model [33]. The fatigue design of the FLS is described in the following sections.

The resulting SCHT dimensions are listed in Table 1, and the symbol definitions are detailed in Fig. 2. The SPPT tower maintained a constant wall thickness, whereas the wall thickness in the steel tower decreased

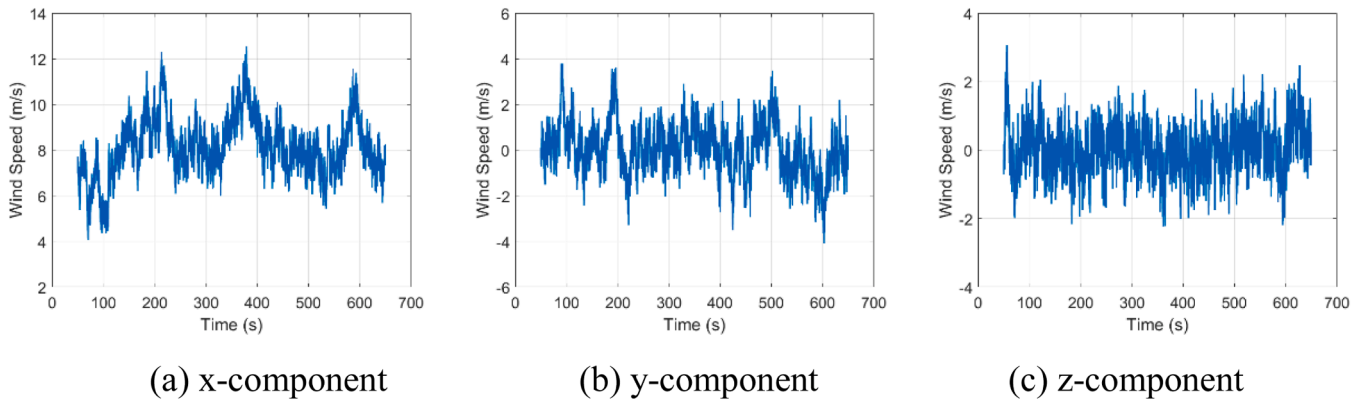


Fig. 7. Wind speed time series at the hub height.

Table 1
Component dimensions of the SCHT.

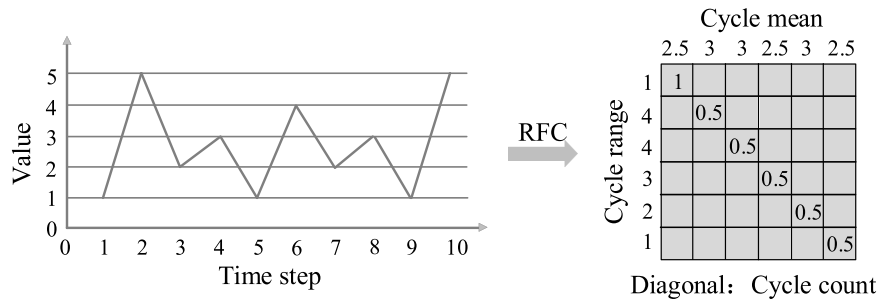
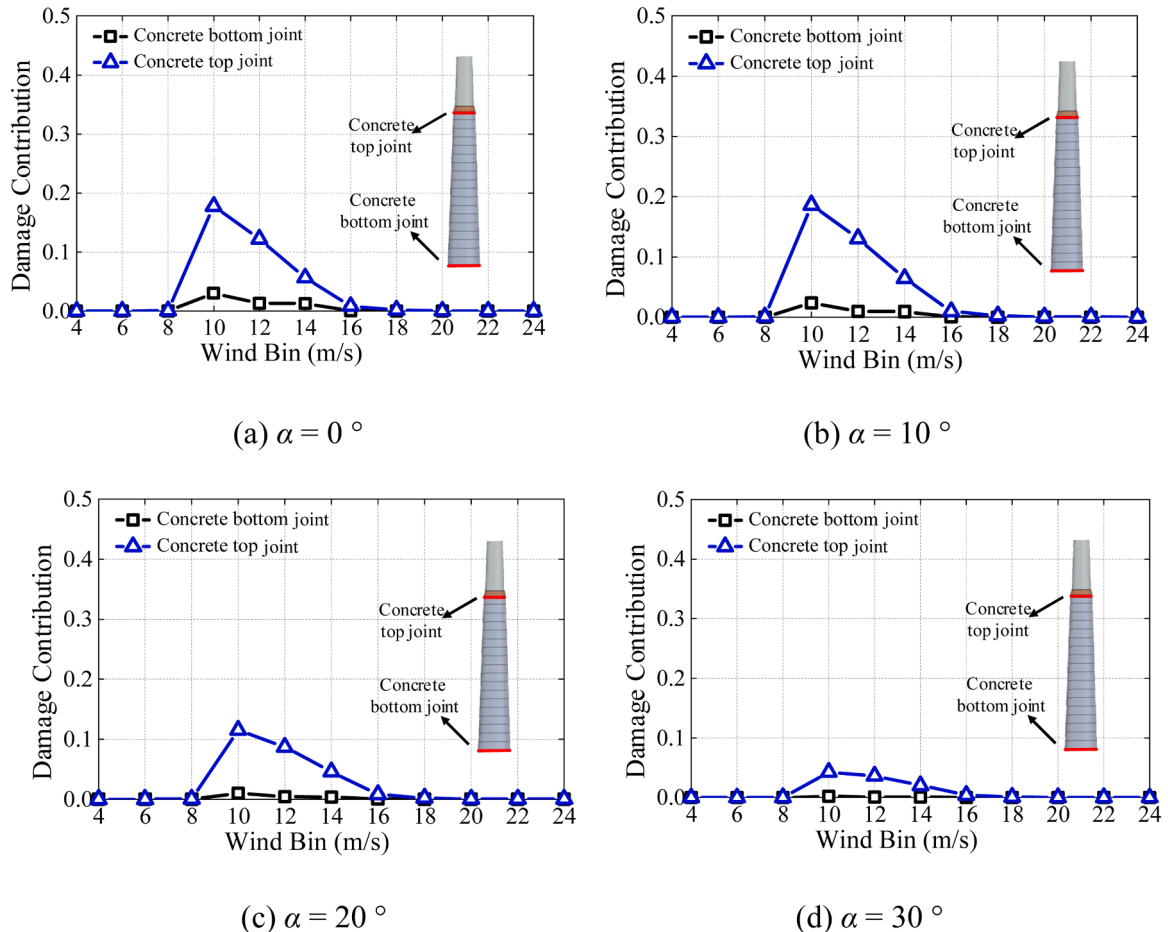
r_{cb}/m	t_c/mm	r_{ct}/m	t_a/mm	r_{sb}/m	t_{sb}/mm	r_{st}/m	t_{st}/mm	l_c/m	F_{pto}/MN
4.60	280	2.50	780	2.15	25.0	1.70	13.0	128.0	80.0

Table 2

First two modal frequencies of the SHT.

Fore-aft direction		Side-to-side direction	
1st	2nd	1st	2nd
0.24 Hz	0.93 Hz	0.24 Hz	0.87 Hz

linearly from bottom to top. Table 2 presents the first two modal frequencies of the SHT in the fore-aft and side-to-side directions. The results in both directions were highly similar, suggesting that variations in the turbine moment of inertia had a minimal impact on the free vibration responses of the SHT.

**Fig. 8.** Rainflow counting matrix obtained from stress time series data.**Fig. 9.** Effects of azimuth angle on damage contributions of different wind bins.

4. Fatigue life calculation

This section explores the influence of the mean stress correction, initial pre-stress, azimuth angle, and sample size on the SPPT tower fatigue life calculation results. The horizontal joints between the tower segments are assumed to be dry joints, with no epoxy or other bonding materials used. Additionally, the fatigue characteristics of the steel tower are provided to emphasise the distinctions between the two tower portions.

Fatigue investigations of SPPT towers involve analysing stress time-series data generated from dynamic simulations and evaluating their results using the fatigue life curves of the adopted materials. The IEC 61400-1 standard [3] recommends performing at least six 10-min stochastic realizations per wind condition. Accordingly, ten simulations per wind bin were conducted in the fatigue analyses, resulting in a total of

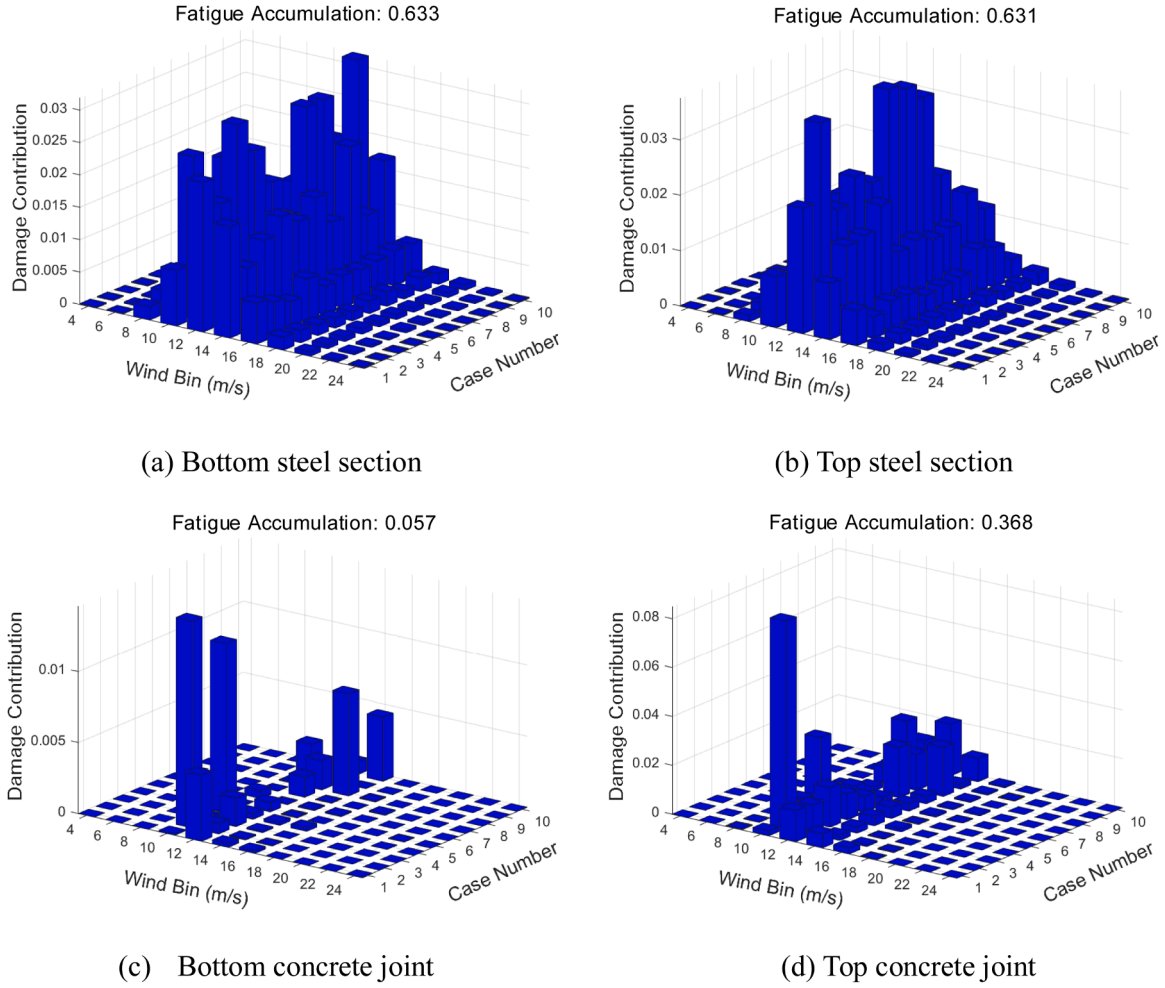


Fig. 10. Variation in damage contribution according to wind bins.

110 simulations for an integrated fatigue evaluation of the 11 wind bins. In each simulation, a unique turbulence seed was used to generate independent random phases at each wind speed. Turbine responses under these wind conditions were simulated using the OpenFAST model described in the previous section.

After obtaining the force components from the OpenFAST model, the sectional stresses of the horizontal joints were theoretically derived by assuming that the plane sections remained planar. The time series local stress $\sigma(t)$ at a specified azimuth angle α is given as:

$$\sigma(t) = \frac{F_z(t) + F_{pt0}}{A_c} + \frac{M_x(t)}{I_{xx}} R \sin \alpha - \frac{M_y(t)}{I_{yy}} R \cos \alpha \quad (1)$$

where F_{pt0} is the vertical component of the initial PT force; $F_z(t)$ is the axial force; $M_x(t)$ and $M_y(t)$ are the fore-aft and side-to-side moments, respectively; A_c is the cross-sectional area; I_{xx} and I_{yy} are the moment of inertia in the fore-aft and side-to-side directions, respectively; R is the outer radius in the horizontal joints; and t is the time of the response time histories.

Existing fatigue-life predictive models are typically based on fatigue tests of specimens under constant-amplitude loading conditions. Therefore, these models cannot directly quantify the fatigue damage caused by the turbine dynamic responses. In this study, the rainflow counting method provided by ASTM [34] was utilised to transform each $\sigma(t)$ into an equivalent rainflow matrix. This matrix includes cycle mean σ_m , cycle range $\Delta\sigma_c$, and cycle count N_c , as illustrated in Fig. 8. The fatigue damage caused by the time series local stress $\sigma(t)$ is then computed via a linear combination of fatigue cycles following the

Palmgren-Miner damage rule [35]. For the fatigue stresses in the SPPT tower, fatigue damage was assessed using the fatigue life model specified in Model Code 2020 [36]. Meanwhile, the fatigue damage to the steel tower was evaluated using the fatigue life model provided by DNV [37].

The fatigue accumulation D^{life} during the design lifetime was estimated as follows:

$$D^{\text{life}} = \sum_{j=1}^{11} D_j \frac{T^{\text{life}} p_j}{t_j} \quad (2)$$

where D_j and t_j are the fatigue damage and total duration of the stress time series data in the j -th wind bin, respectively; T^{life} is the total number of 10-min segments during the design lifetime; p_j is the annual probability of the mean wind speed of the 10-min segments falling in the j -th wind bin. Eq. (2) indicates that the fatigue accumulation D^{life} is calculated by aggregating the damage contribution of each wind bin. Fatigue failure will occur when fatigue accumulation D^{life} exceeded unity.

4.1. Effects of azimuth angle

The fatigue stresses within the tower cross-sections vary owing to the combined fore-aft and side-to-side loads, highlighting the need to identify critical locations along the sectional circumference. The effect of the azimuth angle on damage contributions of different wind bins ($\alpha = 0, 10, 20$, and 30°) is shown in Fig. 9. Owing to the tower's tapered geometry, the fatigue stresses at the base may not consistently govern

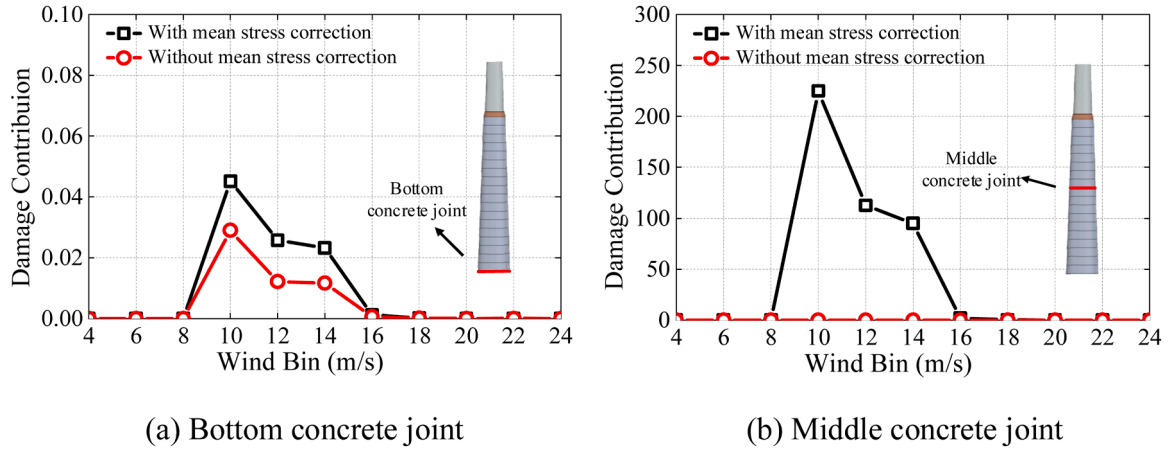


Fig. 11. Damage contributions of different wind bins with and without mean stress correction.

the design criterion. Consequently, the fatigue behaviour at the top and bottom joints of the SPPT tower was compared.

For the bottom joint, fatigue accumulation was very small. Correspondingly, those of the top joint were 0.37, 0.40, 0.26, and 0.11 as the azimuth angle increased from 0° to 30° . This indicates that fatigue accumulation increased up to 10° and then decreased as the azimuth angle continued to increase. The most severe fatigue accumulation occurred near the azimuth angle aligned with the fore-aft direction, indicating the significant contribution of the fore-aft force components to the total fatigue response. Consequently, it is generally reasonable to consider only fore-aft loads ($\alpha = 0^\circ$) when assessing the fatigue accumulation in the concrete portion of a 160-m SHT.

Fig. 10 shows the variation in damage contribution with respect to wind bins, assuming an azimuth angle of zero ($\alpha = 0^\circ$). Ten simulations were conducted for each wind bin. Notably, the fatigue accumulation in different tower portions was significantly affected by wind bins close to a rated wind speed of 11.5 m/s. This is primarily due to these wind bins inducing higher tower loads, resulting in larger mean stress levels at horizontal joints. For the steel tower, the damage contribution exhibited a relatively uniform distribution within same wind bin, whereas for the SPPT tower, it frequently varied across different simulations in the same wind bin. For instance, considering the 10-m/s wind bin at the top joints, the standard deviation of the former was 46 times that of the latter, highlighting the inherent complexity of the long-term fatigue damage of the SPPT tower.

4.2. Effects of mean stress correction

During the fatigue analyses of steel sections, different cycle means are typically transformed to a fixed zero mean using mean stress correction. The fatigue life at an equivalent fixed-mean alternating stress was equal to the fatigue life under alternating stresses with different means. This constant-life transformation is often performed using Marin's general equation [38], as follows:

$$\left(\frac{\sigma_a}{\sigma_f}\right)^n + \left(f \frac{\sigma_m}{\sigma_u}\right)^m = 1 \quad (3)$$

where σ_u is the ultimate strength of the material; σ_a and σ_m are the stress amplitude and mean stress obtained from the response time history, respectively; and σ_f is the zero-mean stress amplitude. In this study, Eq. (3) was used with constants $n = m = f = 1$, which is known as the Goodman model. For the SPPT tower in the SHT, the unbonded PT force between the concrete segments eliminates tensile stress development, rendering the zero-mean alternating stress assumption in the Goodman model unsuitable. Consequently, it was modified as follows to consider nonzero mean stress correction:

$$\sigma_f = \sigma_a \left(\frac{\sigma_u - \sigma_{fm}}{\sigma_u - \sigma_m} \right) \quad (4)$$

where σ_f is the stress amplitude related to the fixed mean σ_{fm} . For horizontal joints in the SPPT tower, σ_{fm} can be specified as the average value

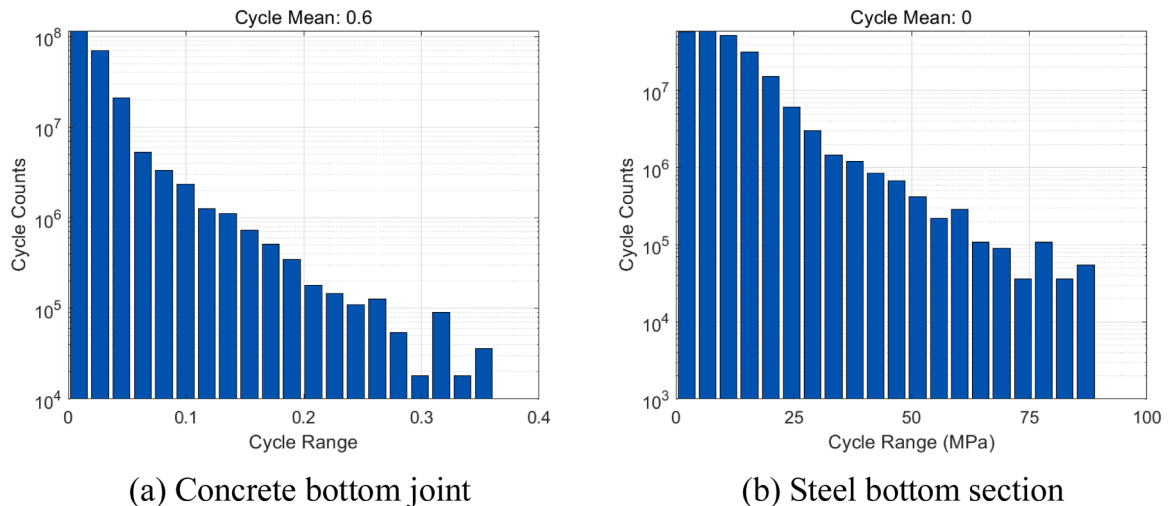


Fig. 12. Rainflow matrices under a wind bin of 10 m/s.

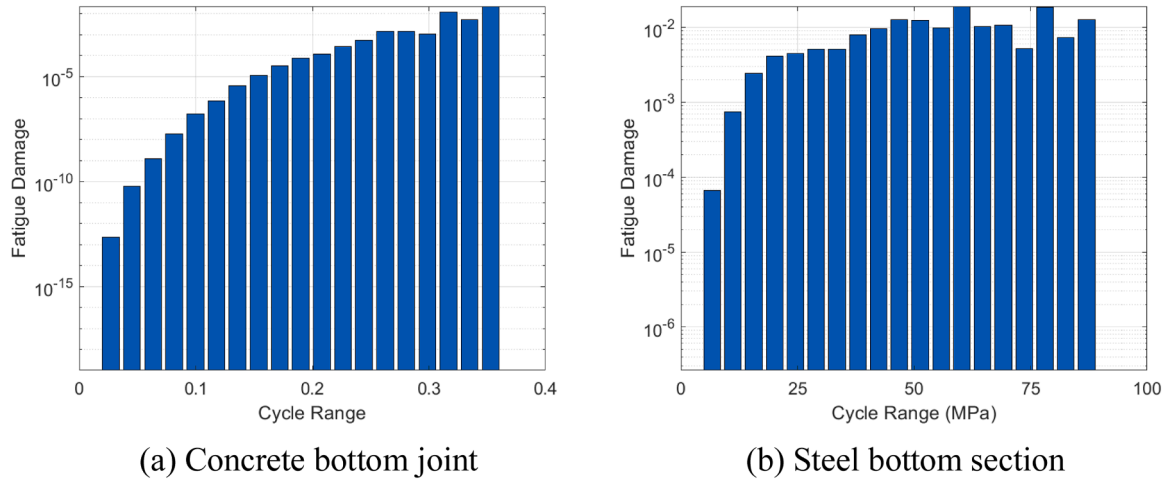


Fig. 13. Fatigue damage under a wind bin of 10 m/s.

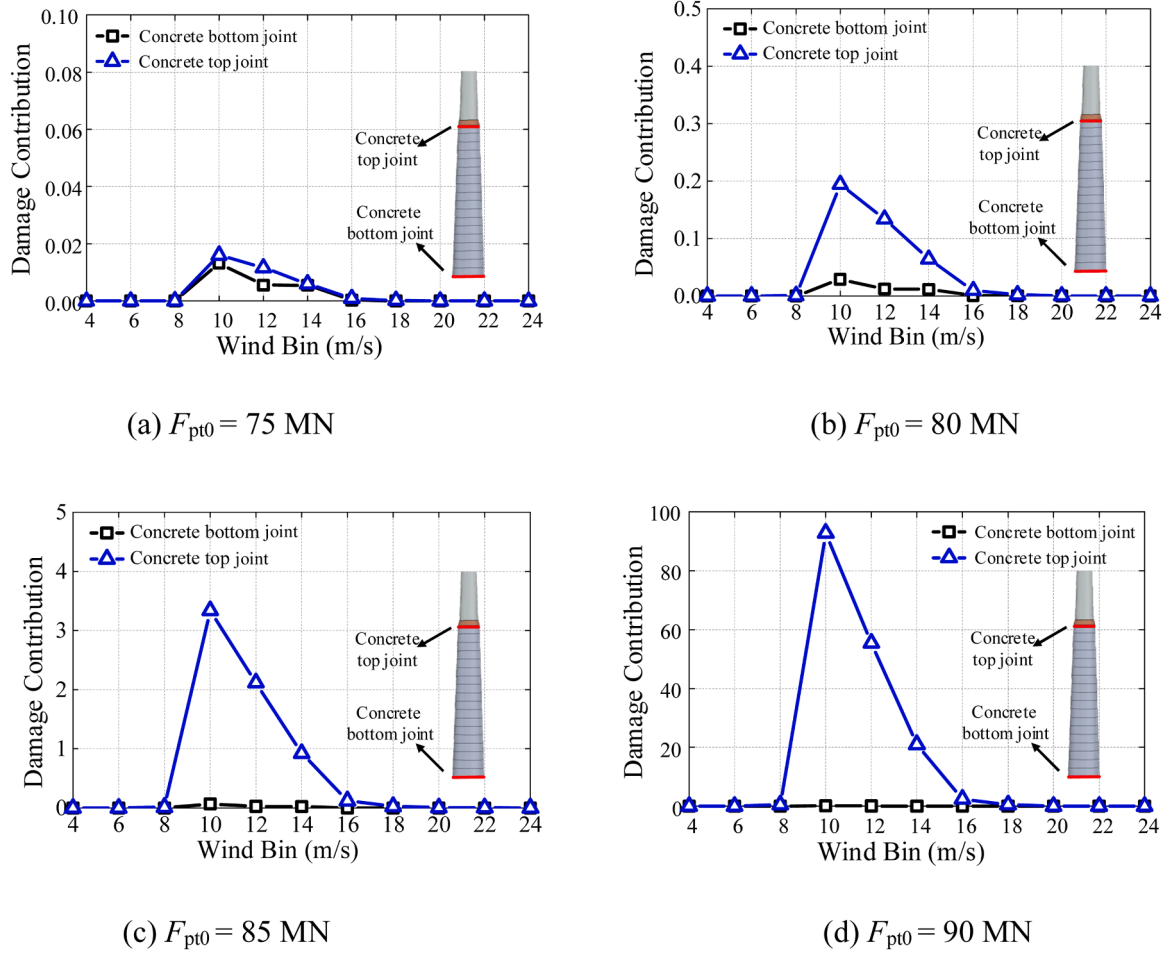


Fig. 14. Effects of initial pre-stress on damage contributions of different wind bins.

of the stress time-series data resulting from the same wind bin.

Fig. 11 shows the damage contribution of different wind bins obtained with and without considering the mean stress correction. It is evident that applying the mean stress correction consistently leads to an overestimation of fatigue accumulation over the design lifetime. Increased damage contribution is prominently observed in the wind bins close to the rated wind speed of 11.5 m/s. The increase in damage contribution in the top joint is not shown because it far exceeds unity.

To elucidate the impact of mean stress correction on fatigue damage, a detailed examination of the fatigue damage associated with each cycle range is presented in Figs. 12 and 13. The cycle range and cycle mean for the concrete are normalized by the design fatigue reference compressive strength from Model Code 2020 [36]. In the rainflow analysis, the cycle range of both the steel and SPPT towers exhibited similar correlations with the cycle counts, as shown in Fig. 12. However, fatigue cycles with larger cycle ranges have smaller cycle counts but exert a greater

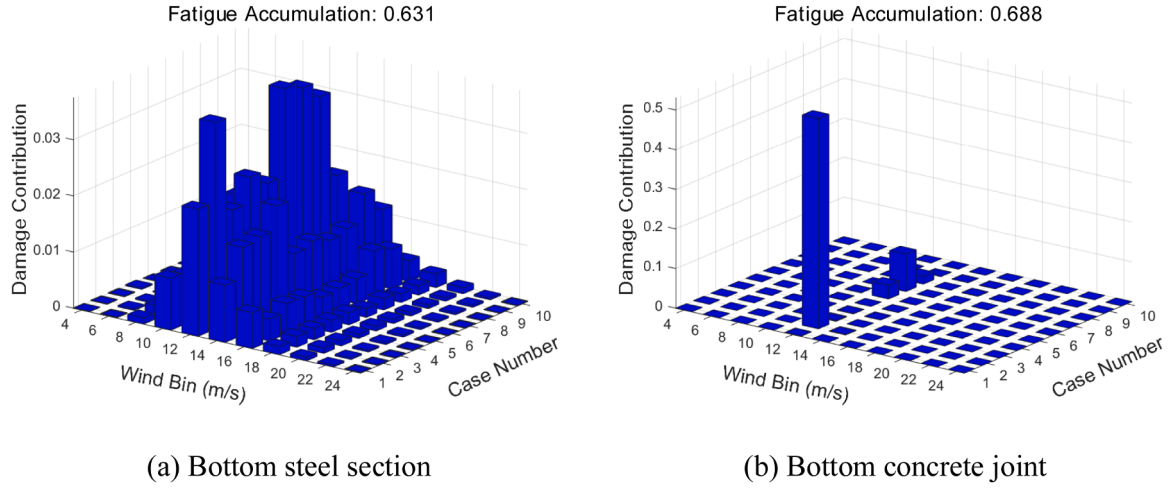


Fig. 15. Variation in damage contribution in the second run.

influence on the fatigue behaviour of the SPPT tower. As shown in Fig. 13a, the fatigue damage exponentially increases (ranging from 10^{-13} to 10^{-2}) over the increasing cycle range. The mean stress correction may widen the difference between the cycle amplitude and mean, thereby inducing a disproportionate alteration in the fatigue life. Fig. 13b shows that the fatigue accumulation in the steel tower is distributed within a narrow range, indicating a relative insensitivity to different cycle ranges. Consequently, mean stress correction was deemed suitable for the steel tower in the SCHT.

4.3. Effects of initial pre-stress

The SPPT tower comprises a stack of precast concrete segments post-tensioned to the foundation using unbonded PT strands and interconnected along the joints without additional reinforcements. The PT strands serve as the primary source of lateral resistance in the SPPT tower. The clamping action of the initial PT force creates a highly rigid initial condition between the concrete segments, resembling securely tied joints.

To investigate the impact of the initial pre-stress, the SPPT tower was analysed using four values ($F_{pt0} = 75, 80, 85,$ and 90 MN). Correspondingly, the design axial ratios at the bottom joint of the SPPT tower varied as 0.20, 0.22, 0.23, and 0.25, while those at the top joint were 0.39, 0.41, 0.44, and 0.46 due to the tapering geometry. Fig. 14 illustrates the fatigue variations of the SPPT tower for different initial pre-stress values. At the bottom joint, the fatigue damage showed a slight

increase with an increase in F_{pt0} across most wind bins and the fatigue accumulation did not exceed unity. However, the damage contributions of different wind bins in the top joint can be significantly affected by increasing F_{pt0} from 80 to 85 and 90 MN. The sharp increase in damage contribution suggests that higher F_{pt0} values noticeably amplify the damage contribution, particularly as the section sizes decrease with tower height. This suggests that progressively increasing the PT force can be detrimental to the tower fatigue behaviour owing to the increasing stress levels.

4.4. Effects of sample size

To assess whether the number of simulations specified by the IEC 61400-1 standard [3] was adequate for the long-term fatigue estimates, the wind turbine loads were recalculated using more independent turbulence seeds in each wind bin. The variability in damage contributions between two separate runs was compared. The results of the first run are shown in Figs. 10a and 10c, while the second run is presented in Fig. 15, with each run consisting of 110 simulations. For the steel tower, small changes (no greater than a 5 % difference) were observed in the different runs. This stable distribution implies that the fatigue load estimates based on the ten simulations are accurate for the steel tower. By contrast, the fatigue damage to the SPPT tower appears significantly susceptible to turbulent wind conditions, as evidenced by the 91.7 % difference observed between the two runs. This considerable variability suggests that ten simulations may not be sufficient to eliminate the

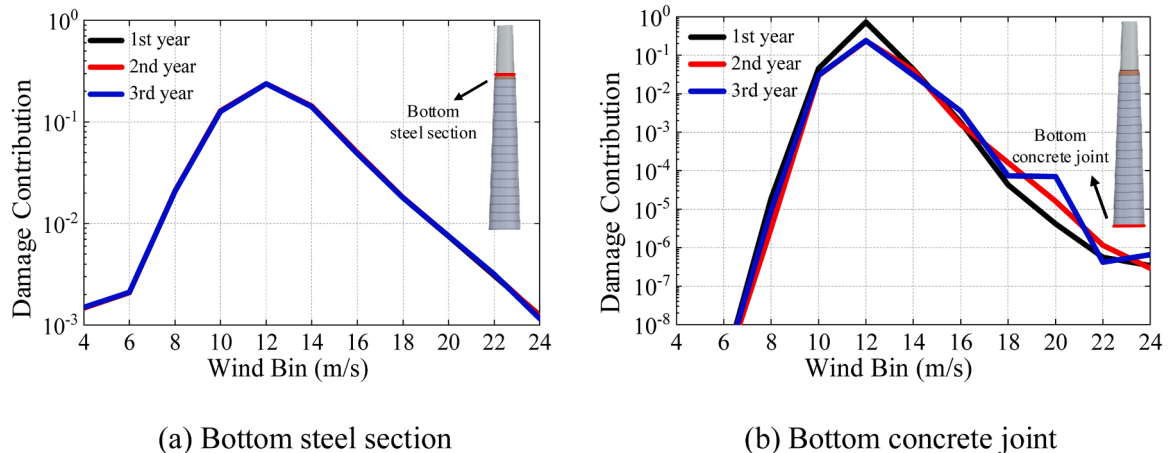


Fig. 16. Damage contributions of different wind bins using year-long simulations.

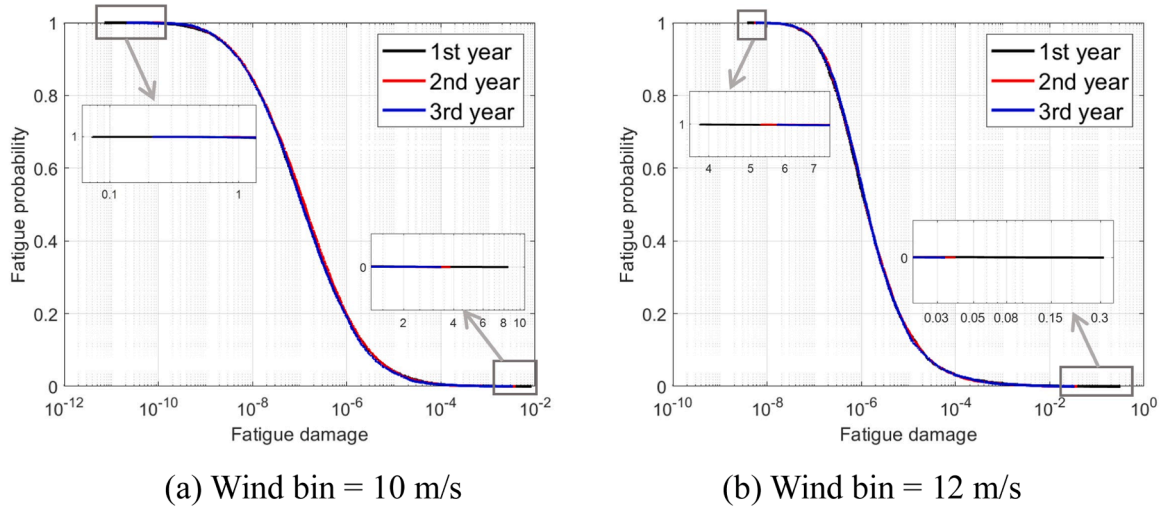


Fig. 17. Fatigue damage at the bottom joint of the SPPT tower.

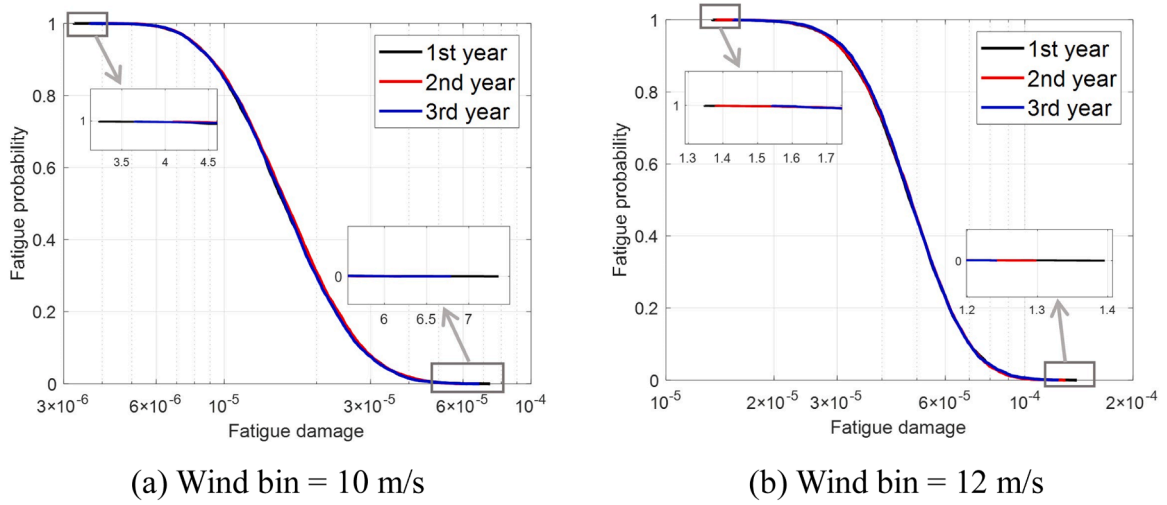


Fig. 18. Fatigue damage at the bottom section of the steel tower.

statistical uncertainty for the SPPT tower.

To assess the long-term damage contribution distribution, three years of turbine operation were simulated, equivalent to $4,6337 \times 3$ fatigue load cases (each representing a 10-minute simulation) after excluding wind speeds below and above the cut-in and -out wind speeds.

As depicted in Fig. 16, the damage contribution from the same wind bin remains almost identical across different one-year simulations for the steel tower. The fatigue accumulation from various year-long datasets was approximately 0.61, closely matching the results from short-term simulations. This indicates that using short-term simulations can

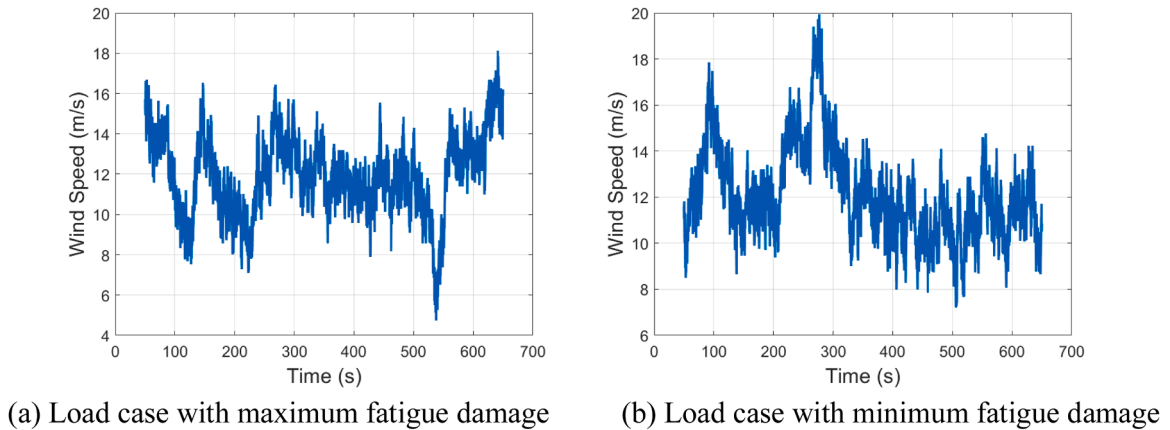
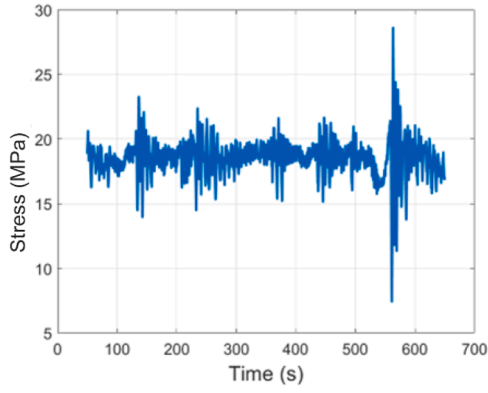
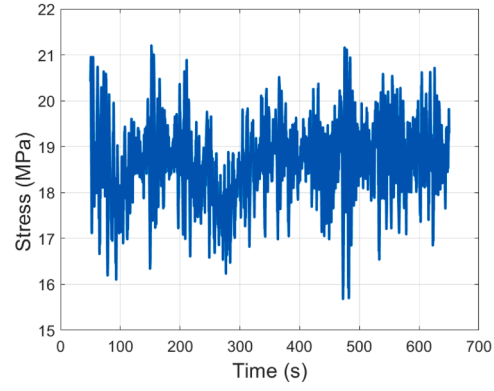


Fig. 19. Wind speed time-series at hub-height.

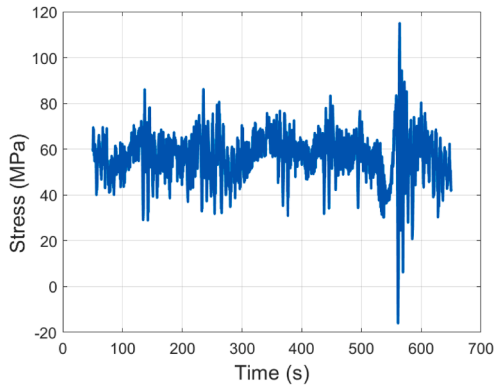


(a) Load case with maximum fatigue damage

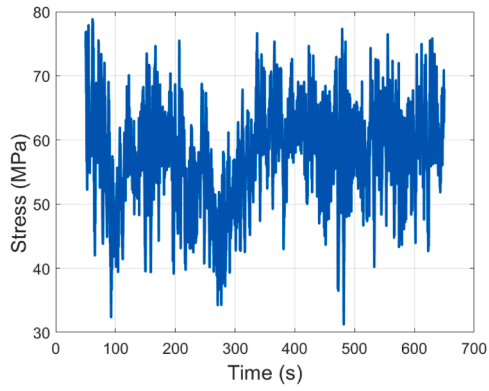


(b) Load case with minimum fatigue damage

Fig. 20. Stress time-series at the bottom joint of the SPPT tower.



(a) Load case with maximum fatigue damage



(b) Load case with minimum fatigue damage

Fig. 21. Stress time-series at the bottom section of the steel tower.

reliably predict the fatigue life of a steel tower in representative wind conditions. Conversely, for the SPPT tower, the fatigue accumulation over the three-year datasets varied significantly, measuring 0.82, 0.31, and 0.31 respectively. It is shown that variability still exists in damage contributions across different year-long simulations, although the differences in fatigue life estimates have decreased to an acceptable level.

Figs. 17 and 18 presents the empirical cumulative distribution (ECD) of fatigue load cases with wind-speed bins of 10 m/s and 12 m/s. Fatigue damage observed in these wind-speed bins contributes most significantly to the overall fatigue accumulation, as they correspond to higher mean stress levels. These results were therefore used to quantify the variability of fatigue damage in the year-long datasets. The ECD reveals a large fatigue damage variability for the SPPT tower, with the ratio of maximum and minimum values reaching approximately 10^8 .

This highlights the fatigue accumulation sensitivity in the SPPT tower to extreme fatigue loads. Conversely, the fatigue damage for the steel tower consistently remained within one order of magnitude under same wind bins. This indicates that the impact of extreme fatigue loads on fatigue accumulation is much smaller for the steel tower.

The stress time histories extracted from the extreme fatigue load cases in the above year-long datasets are highlighted to further illustrate the difference in damage contributions. Both cases shared the same mean wind profile along the tower height. The longitudinal wind speed at the hub height is shown in Fig. 19. The load case with maximum fatigue damage exhibits a significant cycle range, more than four times larger than the load case with minimum fatigue damage, as shown in Figs. 20 and 21.

In these two simulations, the fatigue damage at the steel tower varied

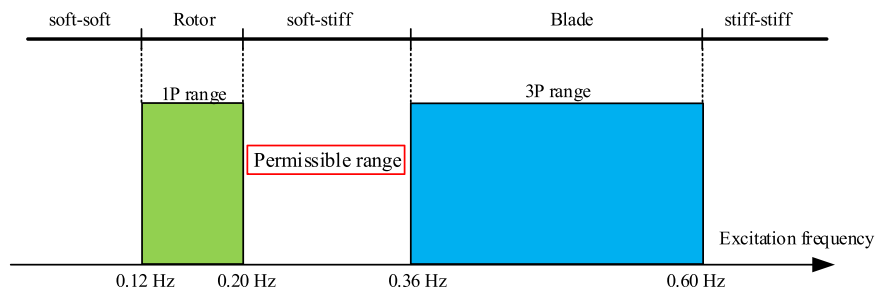


Fig. 22. Permissible range of the tower first modal frequency.

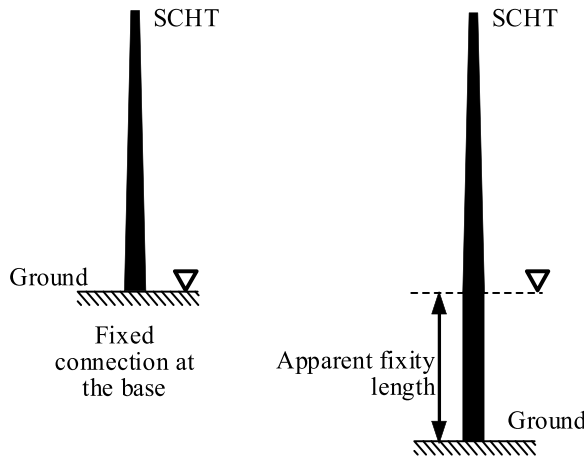


Fig. 23. Conceptual schematics showing details of the fixed-base and flexible foundations.

by approximately 0.1 (i.e., the ratio between maximum and minimum values of fatigue damage), whereas the ratio in the SPPT tower was 10^8 . This suggests that SPPT towers are more susceptible to fatigue cycles with larger mean and range values. As the mean stress level of the SPPT tower increased, the increasing stress amplitude induced a higher rate of change in fatigue damage. Identifying the fatigue load case with maximum fatigue damage is helpful to develop loading protocols and targeted improvements in the future experimental studies.

5. Structural uncertainty analyses

To mitigate the risk of resonance, the natural frequencies of the supporting structures should fall outside the frequencies induced by wind excitation and turbine operation, as shown in Fig. 22. The NREL 5-MW wind turbine typically operates at speeds ranging between 6.9–12.1 revolutions per minute (rpm), corresponding to a rotor operational frequency (1 P) of 0.12–0.20 Hz. Consequently, the blade pass frequency (3 P) was within a range of 0.36–0.60 Hz. Moreover, most of the energy in the wind excitation lies below 1 P. Therefore, it is advisable for the tower frequency f_1 to be situated between the 1 P and 3 P frequencies, as outlined in Table 2.

However, in practice, actual towers may be constructed and erected with deviations from design conditions. These discrepancies may lower the natural frequencies of SCHTs, bringing them closer to turbine operational frequencies. The factors influencing tower dynamic responses inconsistent with the design case can be summarised as follows. a) Elastic modulus of concrete: Previous experimental studies [39,40] revealed a significant discrepancy in the stress-strain relationship of concrete prior to reaching its maximum stress owing to the complex composition of concrete and crack propagation. b) Tower foundation stiffness: The tower's ground restraint [41,42] depends on the strength and stiffness parameters of the soil as well as their interaction with the foundation, making it difficult to achieve the fixed condition assumed in this study. c) Turbine control strategy: Wind turbine loads under different wind speeds can change based on the control approach. The lifespan of the supporting towers can be extended by integrating prognostic-based control schemes into the wind turbine [43–46].

In this section, the impact of these structural uncertainties on the damage contributions of key wind bins is evaluated. The computational model established in the previous sections served as a benchmark. For each assessment, only one parameter was altered, while keeping the other parameters constant. Year-long turbulent wind fields were used to reduce short-term variability effects.

Table 3

Comparison between the first two modal frequencies.

Type	Fore-aft direction		Side-to-side direction	
	1st	2nd	1st	2nd
AF length = 10 m	0.22 Hz	0.85 Hz	0.22 Hz	0.80 Hz
AF length = 20 m	0.20 Hz	0.78 Hz	0.20 Hz	0.74 Hz

5.1. Tower foundation stiffness

In the OpenFAST software [28], the foundation flexibility is typically modelled using the apparent fixity (AF) [41] method owing to its simplicity, as shown in Fig. 23. This involves incorporating a fictitious cantilevered beam beneath the tower base to simulate the soil-foundation interaction. The length and properties of this fictitious beam were carefully defined to match the expected stiffness of the actual soil-foundation system. Table 3 presents the natural frequencies of the AF and fixed-base models. The foundation stiffness in the AF model was selected based on the risk that f_1 approached or slightly overlapped with the turbine operational frequencies.

As shown in Fig. 24, the damage contributions of different wind bins increases significantly as the tower modal frequencies decrease. As f_1 ranged from 0.24 to 0.22 Hz, the damage contributions at the bottom joint for the 10-, 12-, 14-, and 16-m/s wind bins increased 204.8, 139.6, 53.9, and 196.5 %, respectively, compared with the theoretically exact model. Similarly, as f_1 ranged from 0.24 to 0.20 Hz, the damage contributions increased by 433.6, 351.6, 385.5, and 593.9 %, respectively. These results suggest that neglecting the foundation flexibility may significantly underestimate the fatigue accumulation in the SCHT. The damage contributions at the top joint of the SPPT tower exhibited a smaller increase when subjected to the effects of foundation flexibility.

5.2. Elastic modulus of concrete

Various methods are provided in modern building codes for calculating the elastic modulus of concrete. To assess the accuracy of these methods, Noguchi et al. [39] developed a database consisting of more than 3000 uniaxial compression tests. Their findings suggest that the Eurocode-2 equation [31] tends to overestimate the modulus of elasticity of concrete, and most of the residuals (i.e., the difference between the estimations and observations) fall within 0–8000 MPa. A near-mean residual of 4000 MPa was considered to demonstrate the typical deviation effects, corresponding to a 10 % reduction in the design value. As a result, the first two modal frequencies of the SCHT were respectively reduced to 0.21 and 0.82 Hz in the fore-aft direction, and 0.21 and 0.79 Hz in the side-to-side direction.

Fig. 25 shows the variation in damage contributions after considering the reduced elastic modulus. As E_c is decreased by 10 % in the SPPT tower, the damage contributions at the bottom concrete joint varies by 60.1, 66.3, –16.8, and 87.0 % for the 10-, 12-, 14-, and 16-m/s wind bins, respectively. Correspondingly, the damage contributions at the top concrete joint varied by 8.6, 16.7, 0.38, and 20.7 %, respectively. The increased damage is mainly due to the reduction in the elastic modulus of concrete, which in turn increases the likelihood of a resonance condition between the tower and turbine. However, the increase induced by the elastic modulus of concrete was not significant compared to the impact of the tower foundation stiffness.

5.3. Turbine control strategy

Integrating advanced control technology may prevent the premature fatigue failure of wind turbine systems owing to off-design deviations. To achieve this, the tower resonance avoidance mode offered by the ROSCO controller [29] was analysed. This mode enables the avoidance of specific frequency ranges through torque control set points. Its

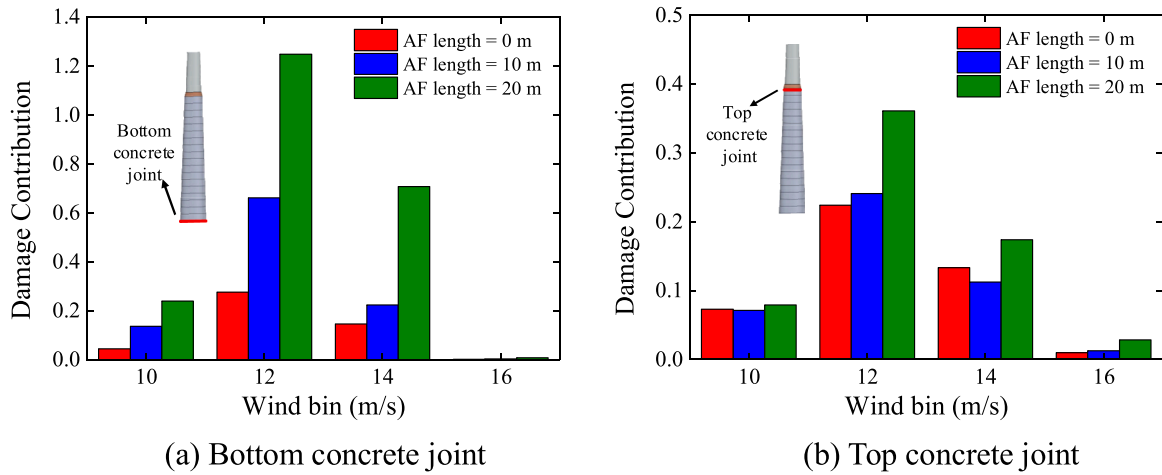


Fig. 24. Damage contribution change according to variations in the tower foundation stiffness.

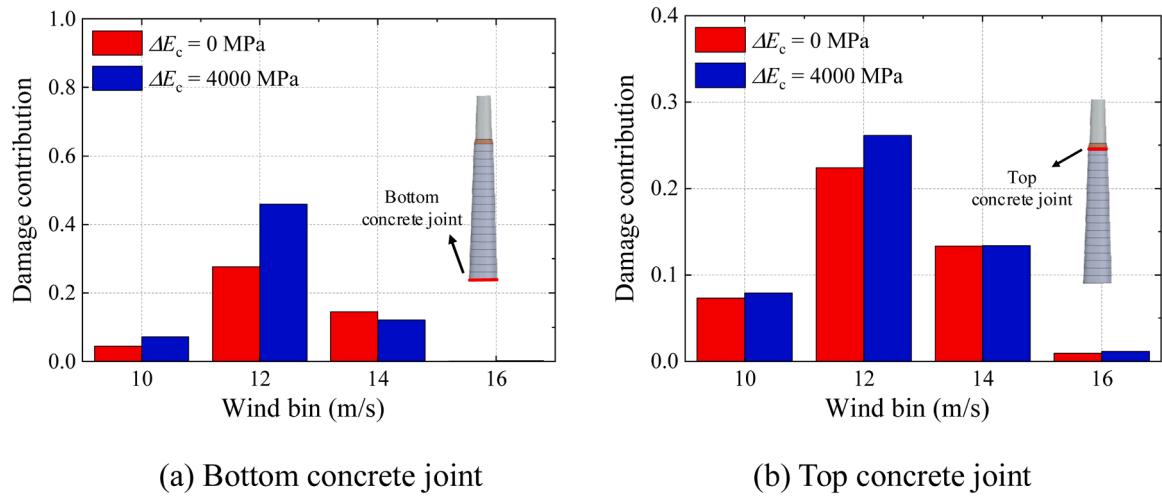


Fig. 25. Damage contribution change according to variations in the elastic modulus concrete.

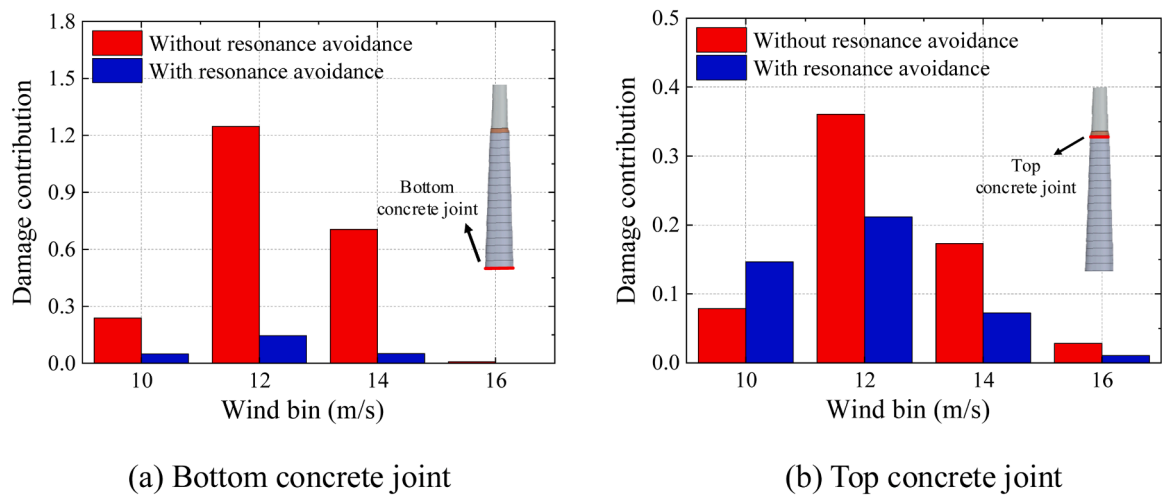


Fig. 26. Damage contribution change according to variations in the turbine control strategy.

effectiveness was examined using the AF model with the lowest deviation frequency ($f_1 = 0.20$ Hz). The torque control set points were changed to avoid a rotor frequency range of 0.18–0.22 Hz.

As shown in Fig. 26, the damage contributions of different wind bins

are significantly reduced by employing the tower resonance avoidance mode. The damage contributions at the bottom concrete joint decreased by 78.9, 88.3, 92.5, and 82.7 % for the 10-, 12-, 14-, and 16-m/s wind bins, respectively, while that of the top concrete joint exhibited similar

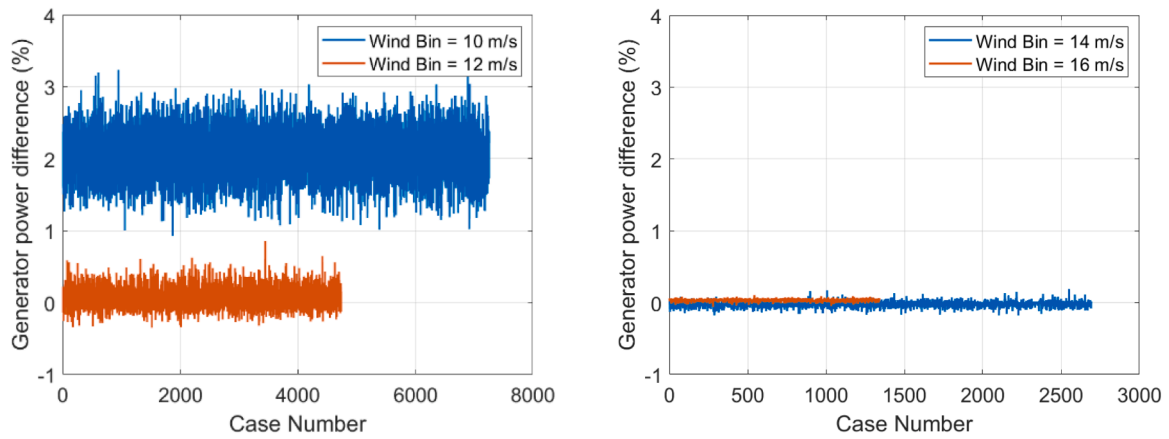


Fig. 27. Generator power change according to variations in the turbine control strategy.

decreases across most wind bins. Notably, the fatigue accumulation of all horizontal joints remained below unity, highlighting the effectiveness of advanced control strategies in mitigating tower fatigue. Additionally, the use of this mode resulted in an average mean output power fluctuation decrease of 2.1, 0.1, 0.0, and 0.0 % for the 10-, 12-, 14-, and 16-m/s wind bins, respectively, as shown in Fig. 27, indicating small impacts on power variation.

6. Conclusion

SPPT towers are typically designed utilized as the lower portion of a SHT. This tower configuration is suitable for wind turbines in regions where achieving higher hub heights is crucial for maximizing wind resource capture. In this study, the fatigue damage of the SPPT tower was investigated using a multi-body dynamic model of a 160-meter SHT. The main dimensions of tower components were determined based on operational limit state loads derived from an NREL 5-MW wind turbine.

The main findings from the fatigue life calculation of SPPT towers are summarized as follows: (i) The mean stress correction methods significantly overestimated the fatigue accumulation in different horizontal joints. In contrast, the average stress correction is more suitable for steel towers. (ii) It is generally reasonable to consider only fore-aft loads when assessing the fatigue accumulation in the concrete portion of a 160-m SHT. (iii) The initial pre-stress had minimal effects on the bottom concrete joint but significantly affected the top concrete joint. (iv) Short-term simulations can accurately predict the fatigue accumulation of steel tower sections; however, a larger sample size is required for horizontal joints in the SPPT tower owing to the variability in fatigue cycles with larger mean and range values.

Structural uncertainty analyses were performed to investigate their effects on the fatigue life of the SPPT towers. Uncertainties in the elastic modulus of concrete and tower foundation stiffness can lead to an underestimation of the tower's natural frequencies, thereby increasing the risk of resonance with the turbine's operating frequencies. Neglecting

these uncertainties can result in a significant underestimation of fatigue accumulation in the SPPT tower. Integrating a tower resonance avoidance mode can prevent the premature fatigue failure of the SHT owing to off-design deviations. However, the additional cost of this control method limits its practical application in the SHT.

The fatigue damage in wind turbine towers accumulates progressively over time as the turbine operates, which may in turn affect the tower's dynamic responses. However, the nonlinearities resulting from accumulated damage cannot be considered in simulation tools like the OpenFast software. Further research is recommended to analyse the impact of these coupled effects on long-term tower performance.

CRediT authorship contribution statement

Huang Xiaogang: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis. **Cui Jie:** Writing – review & editing, Software. **Wang Yu-Hang:** Supervision, Conceptualization. **Li Tian:** Methodology. **Zhou Xuhong:** Supervision, Project administration. **Zhu Dongping:** Writing – review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors gratefully acknowledge the financial support provided by the National Natural Science Foundation of China (52378478), Natural Science Foundation of Chongqing (CSTB2023NSCQ-MSX0650), National Natural Science Foundation Innovation Group Project of China (52221002), and Fundamental Research Funds for the Central Universities of China (2024CDJYXTD-005 and 2024CDJXY021).

Appendix

As stated in IEC 61400-1 [3], the cumulative distribution function of the Rayleigh distribution is defined as,

$$P_R(v) = 1 - \exp \left[-\pi \left(\frac{v}{2V_{ave}} \right)^2 \right] \quad (5)$$

where v is the wind speed. The component power spectral densities used to generate the 3-D wind field are given as,

$$\frac{fS_k(f)}{\sigma_k^2} = \frac{4fL_k/V_{\text{hub}}}{(1 + 6fL_k/V_{\text{hub}})^{5/3}} \quad (6)$$

where f is the frequency in hertz; k is the index referring to the wind speed component ($k = 1$ for the x-component, $k = 2$ for the y-component, and $k = 3$ for the z-component); σ_k and L_k denote the standard deviation and the integral scale parameter of the wind speed component, respectively. Their values are given in Table C.1 of IEC 61400–1 [3].

Data availability

The data that has been used is confidential.

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